PRAVEEN PAL RAWAL

AI/ML Engineer
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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building end-to-end machine learning pipelines, multi-agent LLM systems, and production-grade NLP applications. Proficient in PyTorch, TensorFlow, LangChain, LangGraph, and RAG architectures. Built a 6-agent research platform with retrieval-augmented generation, a self-healing SIEM-grade log processing pipeline (30K+ logs/sec, 93.8% NER accuracy), and real-time voice-enabled AI assistants. Strong foundation in model training & evaluation, data engineering, MLOps, and feature engineering.

TECHNICAL SKILLS

ML/DL Frameworks: PyTorch, TensorFlow, Transformers (Hugging Face), scikit-learn, XGBoost, LightGBM
LLMs & Agents: LangChain, LangGraph, RAG Pipelines, Prompt Engineering, Groq API, OpenAI API, LLaMA-2, Fine-Tuning
NLP & Speech: Natural Language Processing (NLP), Whisper ASR, BERT, FinBERT, spaCy, NLTK, Sentence Transformers, Text Classification, NER
Computer Vision: YOLOv8, BLIP, Grad-CAM, Image Segmentation, Object Detection, MS-COCO
Data & Pipelines: Apache Kafka, Apache Spark, PostgreSQL, MySQL, ChromaDB, Elasticsearch, Redis, Vector Search (HNSW)
MLOps & Deployment: Docker, FastAPI, Streamlit, Gradio, GitHub Actions, CI/CD, REST APIs, MLflow, Model Evaluation
Core Skills: Feature Engineering, Data Preprocessing, Model Training & Optimization, Anomaly Detection, Named Entity Recognition, Statistical Analysis
Languages & Tools: Python (Advanced), SQL (Advanced), Bash, Linux, Git, Jupyter, Power BI

PROFESSIONAL EXPERIENCE

AI Engineering Intern | *AI Core System* Jun 2025 – Aug 2025

- Engineered an end-to-end speech-to-email automation pipeline using Whisper ASR, LangChain, and FastAPI; reduced manual reporting time by ~83% across the organization.
- Implemented silence detection, input validation, and retry logic to handle edge cases in noisy audio environments, achieving <2s end-to-end latency.
- Integrated SendGrid API for automated email dispatch with templated formatting and error handling for 50+ daily automated reports.

AI Workshop Trainer | *NCII SECEX, Rashtriya Raksha University* 2025

- Delivered hands-on applied AI/ML workshops focused on cybersecurity use cases to 150+ professionals including industry practitioners and government officials.
- Designed curriculum covering neural networks, anomaly detection, NLP for threat intelligence, and practical model deployment with FastAPI and Docker.

Data Analytics Intern | *Fashion Trends Analytics, Jaipur* Jan 2023 – Apr 2023

- Built a hybrid recommendation engine combining collaborative filtering and content-based methods; deployed as a demo web application for stakeholder review.
- Automated SQL-based data pipelines and Power BI dashboards, reducing weekly reporting cycle from 5 hours to 2 hours (~60% improvement).
- Analyzed 50K+ customer interaction records to identify purchasing patterns and seasonal trends driving inventory optimization.

Liaison & Operations Coordinator | *All India Police Science Congress (AIPSC)* 2024

- Coordinated end-to-end logistics and operations for a national-level conference with 200+ delegates including senior IPS/IAS officials.

KEY PROJECTS

ResearchSynth AI — Multi-Agent Research Platform — LangGraph, RAG, FastAPI, React 2025

- Architected a 6-agent + critic multi-agent system using LangGraph with conditional feedback loops for autonomous academic research synthesis.
- Built a RAG pipeline with ChromaDB vector store, sentence-transformer embeddings, and Groq API for low-latency LLM inference (<1.5s per query).
- Implemented specialized agents: Search Agent, Reader Agent, Gap Analysis Agent, Experiment Designer, Code Generator, and Writer Agent with a Self-Evaluator critic for quality assurance.
- Deployed full-stack application (FastAPI + React) with real-time SSE streaming, session management, and multi-format export (PDF, Markdown).

AI-Powered Log Processing Pipeline — Kafka, ML, NER, Python, Elasticsearch 2025

- Designed a SIEM-grade, self-healing log normalization pipeline ingesting 30K+ logs/sec from firewalls (Sophos XGS), EDR/XDR, and network monitors via Kafka.
- Achieved 90% log decode rate at 0.18ms/log using regex decoders; remaining 10% gap logs processed by ML pipeline (NER + embeddings) at 15ms/log.
- Built ML-based self-healing decoder using Ollama (phi4-mini) + sentence-transformers; auto-generates new decoder rules with NER accuracy of 93.8% (F1: 0.85).
- Implemented unified 231-field schema mapping, Isolation Forest anomaly detection, threat intelligence enrichment (MITRE ATT&CK), and zero sensitive data exposure via sanitization layer.

Climate-Finance Risk Assessment — PyTorch, TensorFlow, GANs, LLMs 2025

- Modeled multi-modal data (climate indicators, financial time-series, ESG text) using GANs and transformer-based LLMs for probabilistic risk scenario generation.
- Built reproducible ML pipelines with automated validation, missing data imputation, and dataset versioning; generated 2050 climate-risk scenarios for ESG-based investment strategy.

EDUCATION

M.Tech, Data Science & Machine Learning 2024 – 2026
Rashtriya Raksha University (RRU), Gujarat | CGPA: 8.3/10

B.Tech, Computer Science & Engineering (AI) 2019 – 2023
Vivekananda Global University, Jaipur | CGPA: 9.1/10 | Silver Medalist (2nd Rank)

CERTIFICATIONS & ACHIEVEMENTS

- **Neural Networks and Deep Learning** — DeepLearning.AI (Coursera), 2025
- **Big Data Fundamentals** — IBM (Coursera), 2025
- **Runner-up, Smart India Hackathon (National)** — 2022 | Competed among 10,000+ teams nationwide
- **Silver Medalist** — 2nd Rank in B.Tech CSE (AI), Vivekananda Global University, 2023

LEADERSHIP & EXTRACURRICULARS

- **Founder, AI Club (RRU)** — Established university AI club; conducted applied ML sessions for 50–60 students and mentored juniors on training, deployment, and research workflows.
- **Junior Under Officer, NCC (1 Raj Armed SQN)** — Commanded 120 cadets during drills and operations; earned NCC 'C' Certificate with Grade A distinction.